Assignment 3

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Task 1

```
library(terra)
```

```
## terra 1.8.86
```

```
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:terra':
##
##     intersect, union

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
##
## Attaching package: 'tidyr'

## The following object is masked from 'package:terra':
##
##     extract
```

```r
library(ggplot2)
library(lubridate)
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:terra':
##
##     intersect, union

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
states <- st_read("cb_2018_us_state_20m.shp", quiet = TRUE) |>
  filter(!STUSPS %in% c("AK", "HI", "PR")) |>
  st_transform(crs(spei_us))
```

```r
region_map <- c(
  # Northeast
  CT="Northeast", ME="Northeast", MA="Northeast", NH="Northeast", RI="Northeast", VT="Northeast",
  NJ="Northeast", NY="Northeast", PA="Northeast",
  # Midwest
  IL="Midwest", IN="Midwest", MI="Midwest", OH="Midwest", WI="Midwest",
  IA="Midwest", KS="Midwest", MN="Midwest", MO="Midwest", NE="Midwest", ND="Midwest", SD="Midwest",
  # South
  DE="South", FL="South", GA="South", MD="South", NC="South", SC="South", VA="South", WV="South", DC="S
  AL="South", KY="South", MS="South", TN="South",
  AR="South", LA="South", OK="South", TX="South",
  # West (CONUS)
  AZ="West", CO="West", ID="West", MT="West", NV="West", NM="West", UT="West", WY="West",
  CA="West", OR="West", WA="West"
)

states$region <- unname(region_map[states$STUSPS])

regions <- states |>
  filter(!is.na(region)) |>
  group_by(region) |>
  summarise(geometry = st_union(geometry), .groups = "drop")

regions_v <- vect(regions)
```

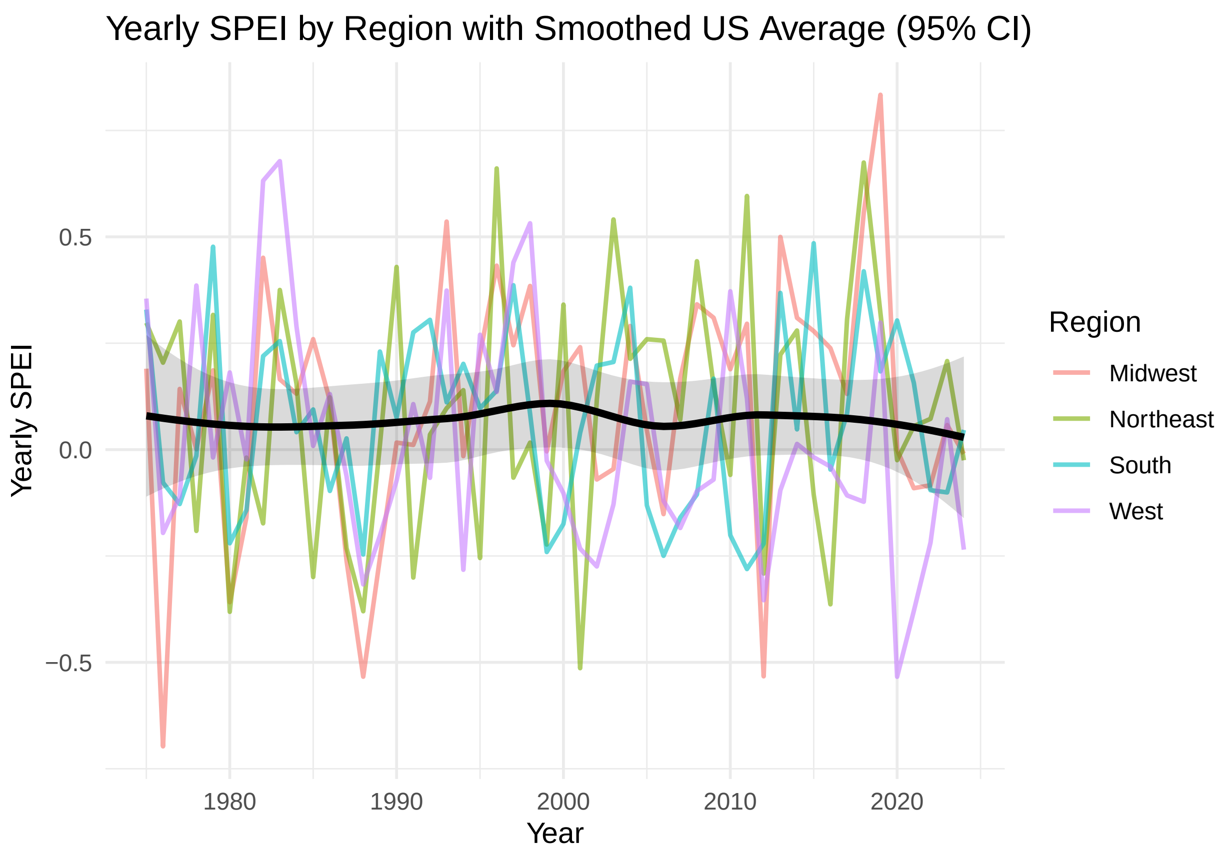
```r
# Area-weighted mean SPEI by region
reg_ex <- terra::extract(spei_us, regions_v, fun = mean, weights = TRUE, na.rm = TRUE)

# Build panel: region x date x spei
panel <- reg_ex |>
  as.data.frame() |>
  mutate(region = regions$region[ID]) |>
  pivot_longer(
```

```r
    cols = -c(ID, region),
    names_to = "date",
    values_to = "spei"
  ) |>
  mutate(date = as.Date(date)) |>
  select(region, date, spei) |>
  arrange(region, date)

# Yearly SPEI per region
panel_year <- panel |>
  mutate(year = year(date)) |>
  group_by(region, year) |>
  summarise(spei = mean(spei, na.rm = TRUE), .groups = "drop")
```

```r
avg_year <- panel_year |>
  group_by(year) |>
  summarise(us_mean = mean(spei, na.rm = TRUE), .groups = "drop")
```

```r
ggplot() +

  # Regional yearly lines (no smooth, no CI)
  geom_line(
    data = panel_year,
    aes(x = year, y = spei, color = region, group = region),
    alpha = 0.6,
    linewidth = 0.8
  ) +


  # US average smooth WITH confidence interval ribbon (transparent)
  geom_smooth(
    data = avg_year,
    aes(x = year, y = us_mean),
    method = "loess",        # or "gam" if you prefer
    se = TRUE,
    color = "black",
    fill = "black",
    alpha = 0.15,
    linewidth = 1.3
  ) +

  labs(
    x = "Year",
    y = "Yearly SPEI",
    color = "Region",
    title = "Yearly SPEI by Region with Smoothed US Average (95% CI)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Task 2

```r
library(sf)
library(terra)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats 1.0.1     v stringr 1.5.2
## v purrr   1.1.0     v tibble  3.3.0
## v readr   2.1.6
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x tidyr::extract() masks terra::extract()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(gdistance)
```

```
## Loading required package: raster
## Loading required package: sp
##
## Attaching package: 'raster'
```

```
##
## The following object is masked from 'package:dplyr':
##
##     select
##
## Loading required package: igraph
##
## Attaching package: 'igraph'
##
## The following object is masked from 'package:raster':
##
##     union
##
## The following objects are masked from 'package:purrr':
##
##     compose, simplify
##
## The following object is masked from 'package:tibble':
##
##     as_data_frame
##
## The following objects are masked from 'package:lubridate':
##
##     %--%, union
##
## The following object is masked from 'package:tidyr':
##
##     crossing
##
## The following objects are masked from 'package:dplyr':
##
##     as_data_frame, groups, union
##
## The following objects are masked from 'package:terra':
##
##     blocks, compare, union
##
## The following objects are masked from 'package:stats':
##
##     decompose, spectrum
##
## The following object is masked from 'package:base':
##
##     union
##
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
##
```

```
## Attaching package: 'gdistance'
##
## The following object is masked from 'package:igraph':
##
##     normalize
```

```r
library(raster)
```

1. We load the ne_10m_populated_places shapefile and filter to Spain. We then rank cities by population and keep the top 10.

```r
# load country boundaries
countries <- st_read("data/ne_10m_admin_0_countries.shp")
```

```
## Reading layer `ne_10m_admin_0_countries' from data source
##   `/Users/yaz/Desktop/BSE/Term 2/Geospatial DS/Assignment 3/data/ne_10m_admin_0_countries.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 258 features and 168 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: -180 ymin: -90 xmax: 180 ymax: 83.6341
## Geodetic CRS:  WGS 84
```

```r
# extract Spain polygon
spain <- countries %>% filter(ADMIN == "Spain")
# load populated places
places <- st_read("data/ne_10m_populated_places.shp")
```

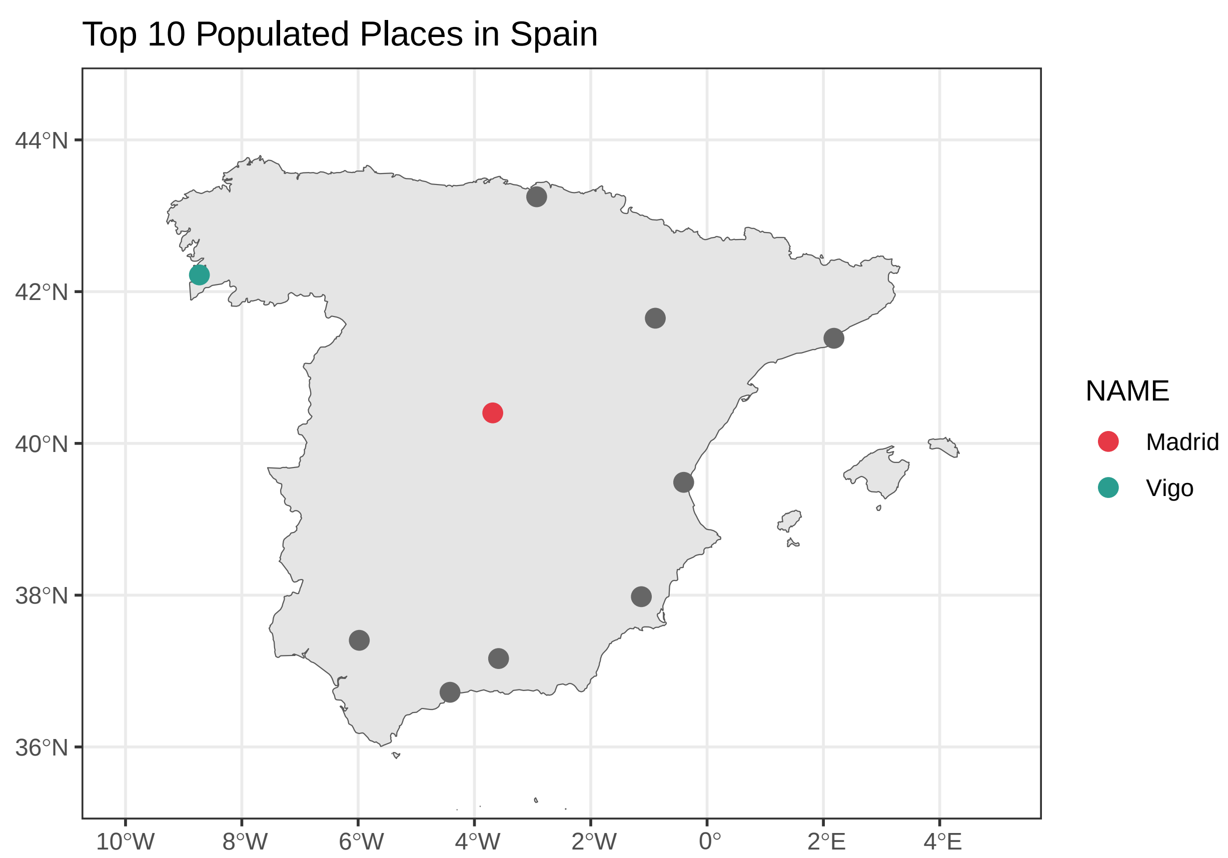
```
## Reading layer `ne_10m_populated_places' from data source
##   `/Users/yaz/Desktop/BSE/Term 2/Geospatial DS/Assignment 3/data/ne_10m_populated_places.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 7342 features and 137 fields
## Geometry type: POINT
## Dimension:     XY
## Bounding box:  xmin: -179.59 ymin: -90 xmax: 179.3833 ymax: 82.48332
## Geodetic CRS:  WGS 84
```

```r
# make sure CRS matches
places <- st_transform(places, st_crs(spain))
# filter places within Spain
places_spain <- st_filter(places, spain)
# get top 10 by population
top10 <- places_spain %>%
  arrange(desc(POP_MAX)) %>%
  slice(1:10)
print(top10 %>% st_drop_geometry() %>% dplyr::select(NAME, POP_MAX))
```

```
##         NAME POP_MAX
## 1     Madrid 5567000
## 2  Barcelona 4920000
## 3    Seville 1212045
```

```
## 4     Bilbao  875552
## 5   Valencia  808000
## 6   Zaragoza  649404
## 7     Málaga  550058
## 8     Murcia  406807
## 9    Granada  388290
## 10      Vigo  378952
```

```
# plot
ggplot() +
  geom_sf(data = spain) +
  geom_sf(data = top10 %>% filter(!NAME %in% c("Madrid", "Vigo")),
          color = "gray40", size = 3) +
  geom_sf(data = top10 %>% filter(NAME %in% c("Madrid", "Vigo")),
          aes(color = NAME), size = 3) +
  scale_color_manual(values = c("Madrid" = "#e63946", "Vigo" = "#2a9d8f")) +
  labs(title = "Top 10 Populated Places in Spain", color = "NAME") +
  theme_bw() +
coord_sf(xlim = c(-10, 5), ylim = c(35.5, 44.5))
```



2. We load the gROADS-v1-europe road network and crop it to Spain's boundary using st_intersection().

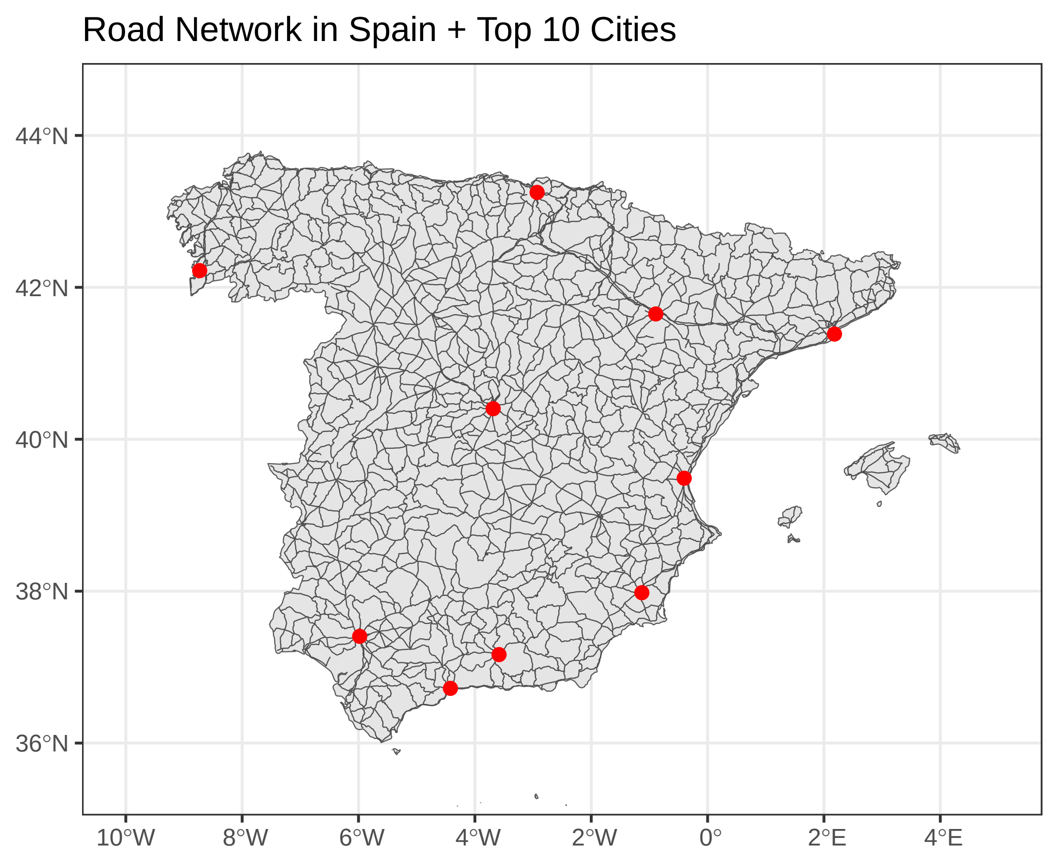
```
# load gROADS Europe shapefile
roads_europe <- st_read("data/gROADS-v1-europe.shp")
```

```
## Reading layer `gROADS-v1-europe' from data source
##   `/Users/yaz/Desktop/BSE/Term 2/Geospatial DS/Assignment 3/data/gROADS-v1-europe.shp'
##   using driver `ESRI Shapefile'
## Simple feature collection with 509757 features and 36 fields
## Geometry type: MULTILINESTRING
## Dimension:     XY
## Bounding box:  xmin: -74 ymin: 27 xmax: 48 ymax: 77.22398
## Geodetic CRS:  WGS 84
```

```r
# reproject to match Spain
roads_europe <- st_transform(roads_europe, st_crs(spain))

# crop roads to Spain
roads_spain <- st_intersection(roads_europe, spain)
```

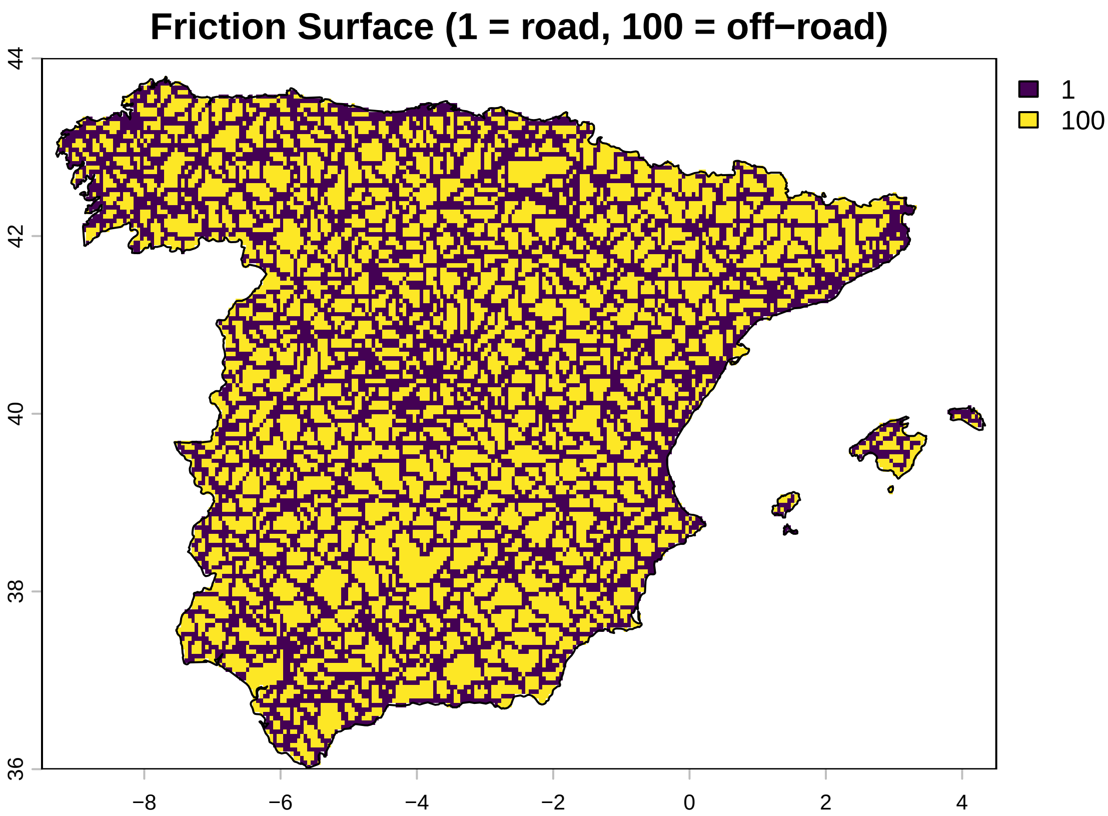
```
## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```

```r
# plot
ggplot() +
  geom_sf(data = spain) +
  geom_sf(data = roads_spain, linewidth = 0.2, color = "gray30") +
  geom_sf(data = top10, color = "red", size = 2) +
  labs(title = "Road Network in Spain + Top 10 Cities") +
  theme_bw() +
coord_sf(xlim = c(-10, 5), ylim = c(35.5, 44.5))
```



3. We build a friction surface by rasterizing the road network. Road cells get a cost of 1 and off-road cells get a cost of 100. We then build a transition matrix and compute the full 10x10 bilateral distance matrix between all city pairs.

```r
# convert Spain to terra vector
spain_vect <- vect(spain)

# create raster template over Spain at 0.05 degree resolution
r_template <- rast(spain_vect, res = 0.05)

# rasterize roads: cells with roads = 1
roads_rast <- rasterize(vect(roads_spain), r_template, field = 1)

# assign friction: 1 on-road, 100 off-road
fv <- values(roads_rast)
fv[is.na(fv)] <- 100
fv[fv == 1]   <- 1
values(roads_rast) <- fv

# mask to Spain only
spain_mask <- rasterize(vect(spain), r_template, field = 1)
roads_rast <- mask(roads_rast, spain_mask)

# plot
ext_spain <- c(-9.5, 4.5, 36, 44)
plot(roads_rast, main = "Friction Surface (1 = road, 100 = off-road)", ext = ext_spain)
plot(spain_vect, add = TRUE)
```




```r
# convert to old raster format (required by gdistance)
r_raster <- raster::raster(roads_rast)

# build transition matrix (8 directions, mean function)
tr_matrix <- transition(
  x                  = r_raster,
  transitionFunction = mean,
  directions         = 8
)

# geocorrect for diagonal distances
tr_matrix <- geoCorrection(tr_matrix, type = "c")
```

```r
# compute full 10x10 cost-distance matrix
dist_friction <- costDistance(
  tr_matrix,
  as(top10, "Spatial")
) %>% as.matrix()

rownames(dist_friction) <- top10$NAME
colnames(dist_friction) <- top10$NAME

# display matrix
print(round(dist_friction, 0))
```

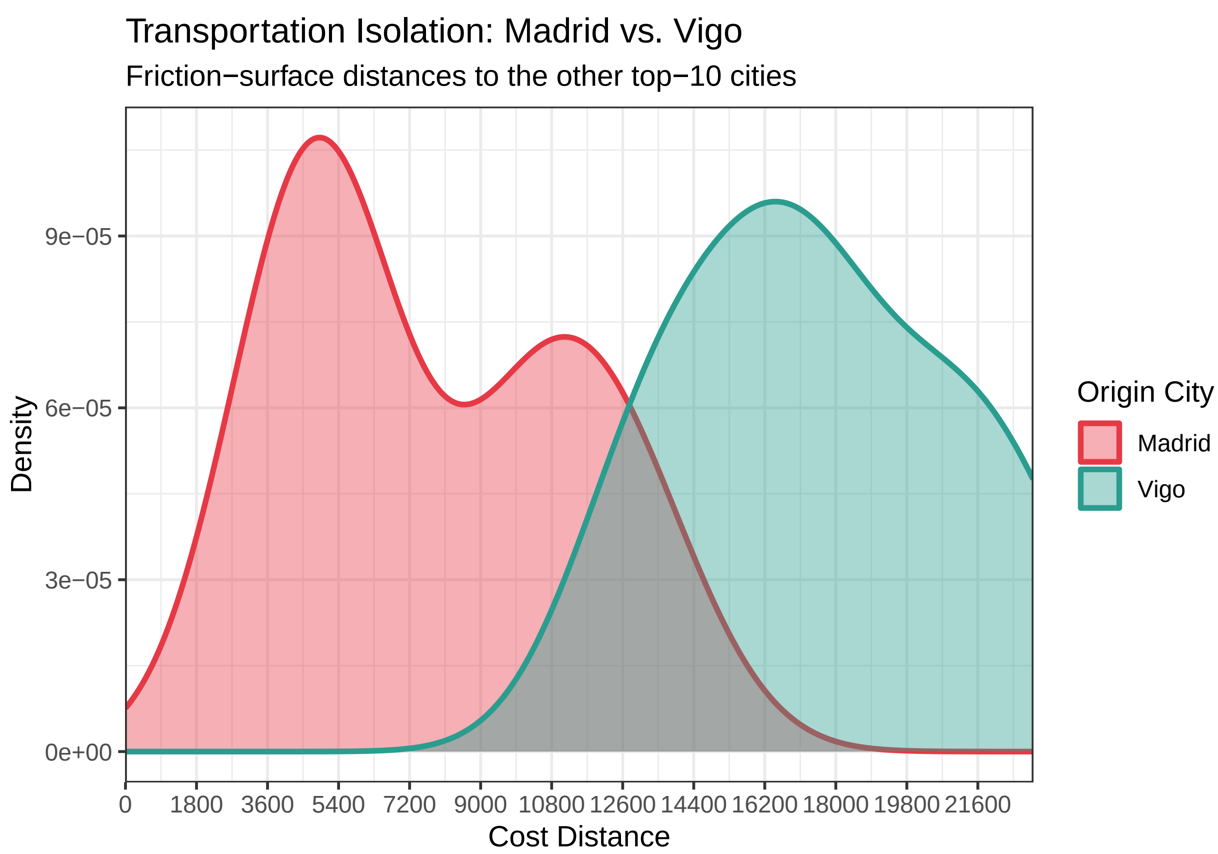
```
##             Madrid Barcelona Seville Bilbao Valencia Zaragoza Málaga Murcia
```

```
## Madrid          0      11958    9991   4851      4459      4435  10365   5150
## Barcelona   11958          0   20158  11133      9850      8017  19990  11893
## Seville      9991      20158       0  13995     11677     13178  11708  10919
## Bilbao       4851      11133   13995      0      6861      3923  14496   8428
## Valencia     4459       9850   11677   6861         0      4140  11425   2762
## Zaragoza     4435       8017   13178   3923      4140         0  13347   6183
## Málaga      10365      19990   11708  14496     11425     13347      0   9264
## Murcia       5150      11893   10919   8428      2762      6183   9264      0
## Granada      5221      14089    8059   9210      5403      7718   6404   3212
## Vigo        13173      22938   20105  13133     17028     15247  21467  17799
##            Granada  Vigo
## Madrid        5221 13173
## Barcelona    14089 22938
## Seville       8059 20105
## Bilbao        9210 13133
## Valencia      5403 17028
## Zaragoza      7718 15247
## Málaga        6404 21467
## Murcia        3212 17799
## Granada          0 16608
## Vigo         16608     0
```

4. We compare cost distances from Madrid and Vigo to all other cities using geom_density(). The city with the higher mean distance is more isolated.

```r
# extract distances from Madrid and Vigo to all other cities
madrid_idx <- which(rownames(dist_friction) == "Madrid")
vigo_idx   <- which(rownames(dist_friction) == "Vigo")

dist_df <- bind_rows(
  tibble(city = "Madrid", distance = dist_friction[madrid_idx, -madrid_idx]),
  tibble(city = "Vigo",   distance = dist_friction[vigo_idx,   -vigo_idx])
)

# dynamic x-axis upper limit
max_val <- ceiling(max(dist_df$distance) / 100) * 100
# density plot
ggplot(dist_df, aes(x = distance, fill = city, color = city)) +
  geom_density(alpha = 0.4, linewidth = 1) +
  scale_fill_manual(values = c("Madrid" = "#e63946",
                               "Vigo"   = "#2a9d8f")) +
  scale_color_manual(values = c("Madrid" = "#e63946",
                                "Vigo"   = "#2a9d8f")) +
  scale_x_continuous(
    limits = c(0, max_val),
    breaks = seq(0, max_val, by = 1800),
    expand = c(0, 0)
  ) +
  labs(
    title    = "Transportation Isolation: Madrid vs. Vigo",
    subtitle = "Friction-surface distances to the other top-10 cities",
    x        = "Cost Distance",
    y        = "Density",
    fill     = "Origin City",
```

```r
    color     = "Origin City"
  ) +
  theme_bw()
```




```r
# conclusion
m_madrid <- round(mean(dist_friction[madrid_idx, -madrid_idx]), 1)
m_vigo   <- round(mean(dist_friction[vigo_idx, -vigo_idx]), 1)

cat("Mean distance from Madrid:", m_madrid, "\n")
```

```
## Mean distance from Madrid: 7733.6
```

```r
cat("Mean distance from Vigo:", m_vigo, "\n\n")
```

```
## Mean distance from Vigo: 17499.8
```

```r
cat("Vigo has a higher average friction-surface distance to the other top-10 cities,\n",
    "indicating greater transportation isolation relative to Madrid.")
```

```
## Vigo has a higher average friction-surface distance to the other top-10 cities,
##  indicating greater transportation isolation relative to Madrid.
```